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DEMOGRAPHIC HISTORY OF POLOTSK, TUROV AND BELARUSIAN RUS (10000BC - 1300AD).

Regional Boundaries, Devolved Principalities and Population Estimates in North-Western Rus.

8 November 2026 - Aust Kätzchen | austkaeztchen@confoederatio.org

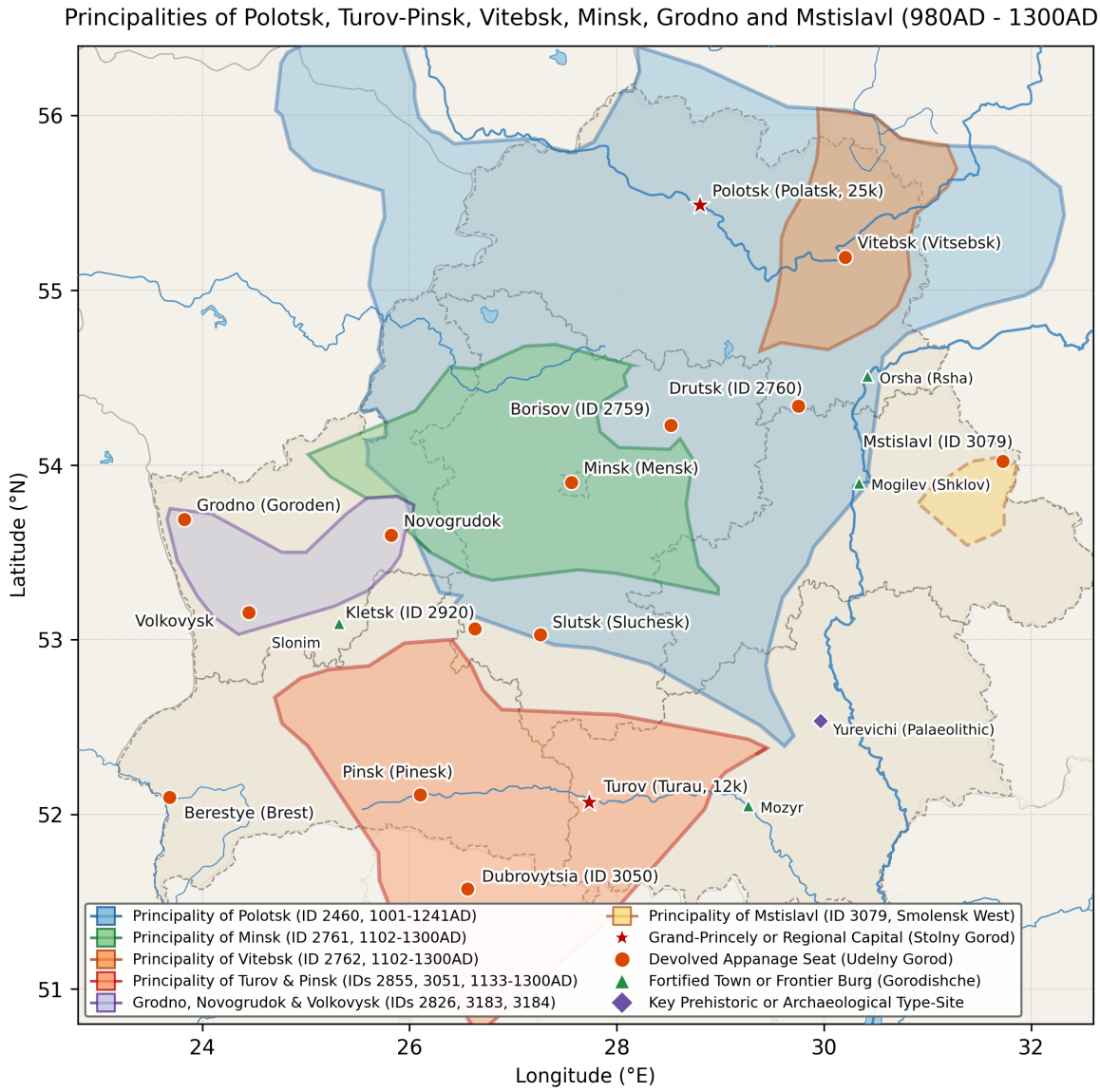


Figure 1. Regional Boundaries and Fortified Settlement Hierarchy of Polotsk, Vitebsk, Minsk, Turov, Pinsk and Black Ruthenia (10000BC - 1300AD).

Executive Summary and Geographic Scope

This literature review establishes the Holocene, Iron Age, Kievan Rus and immediate post-Mongol demographic series (10000BC - 1300AD) together with the fortified urban settlement hierarchy for Region 4: North-Western Rus (Polotsk, Turov and Belarusian Rus) within the Kievan Rus macro-region. In the SVEA spatial partition, this region encompasses the 207 600 km^2 basin network of modern Belarus (the Upper Dnieper, Berezina, Sozh, Pripyat, Western Dvina, Upper Neman and Western Bug-Mukhavets watersheds), alongside the historical Polotsk tributary principalities of the lower Western Dvina (*Jersika (*Gerzike*)* and *Koknese (*Kukenois*)* in Latgale and Selonia) and the southern Pripyat-Horyn Polesian appanage of *Dubrovytsia* and *Stepan*.

Strictly book-ended at the Mongol Conquests (1238-1241AD) and the immediate post-conquest devolved principalities (1240AD - 1300AD), this regional synthesis harmonises directly with the northern and eastern European Russia synthesis (specifically *Novgorod-Pskov* [310,0k at 1000AD; 480,0k at 1200AD] and *Smolensk and Upper Dnieper* east of 31,0°E [220,0k at 1000AD; 340,0k at 1200AD]). Across the Kievan Rus and appanage eras (980AD - 1300AD), North-Western Rus comprised five interconnected dynastic and hydrographic clusters [1][2][3][4]:

1. **The Polotsk-Vitebsk and Lower Dvina Principalities** (polity IDs 2460, 2762, 2459, 3106): The senior Izyaslavichi patrimony along the Western Dvina (Daugava) river, encompassing the core Principality of Polotsk (2460, 1001-1241AD), the Principality of Vitebsk (2762, 1102-1321AD) with *Orsha* and *Kopys* and the lower Dvina tributary principalities of Jersika (2459, 1001-1210AD) and Koknese (3106, 1191-1210AD) [5][6][7][8].
2. **The Southern Izyaslavichi Principalities of Minsk, Drutsk and Borisov** (polity IDs 2761, 2760, 2759): Spanning the Minsk Upland, the Upper Berezina, Svisloch and Upper Drut portages, comprising the Principality of Minsk (2761, 1102-1327AD) (relocated from the 10th-11th century *Menka* complex to the Svisloch-Nemiga *Zamchishche* after 1067AD), the Principality of Drutsk (2760, 1102-1251AD) and the Principality of Borisov (2759, 1102-1301AD), alongside *Zaslavl (*Izyaslavl*)*, *Logoisk (*Logozhsk*)* and *Svisloch-on-Berezina* [1][9].
3. **The Turov-Pinsk Principalities and Berestye Land in Pripyat Polesia** (polity IDs 2855, 3356, 3471, 3051, 2920, 3050): The Polesian alluvial corridor of the Pripyat, Pina, Yaselda, Sluch, Lan, Ptich and Horyn rivers, comprising the Principality of Turov (2855, 3356, 3471, 1133-1341AD), the Principality of Pinsk (3051, 1175-1321AD), the Principality of Kletsk (2920, 1148-1401AD) with *Slutsk* and *Kopyl*, the Principality of Dubrovytsia (3050, 1175-1301AD) and the Western Bug-Mukhavets March of Berestye (*Brest*) and Kamenets [3][10][11].
4. **Black Ruthenia** (Upper Neman) (*Chornaya Rus*, polity IDs 2826, 3183, 3184): The Slavic-Baltic contact zone along the Upper Neman, Shchara and Rossa rivers, anchored by the Principality of Grodno (*Goroden*) (2826, 1128-1201AD), the Principality of Novogrudok (3183, 1201-1241AD) and the Principality of Volkovysk (3184, 1201-1241AD) with *Slonim*, *Zditovo-on-Molchad* and *Turiisk-on-Neman* [12][13][14].
5. **The Mstislavl, Western Smolensk and Upper Sozh Corridor** (polity ID 3079): The Principality of Mstislavl (3079, 1181-1198AD) on the Vikhra river, together with the western Smolensk and northern Chernigov-Radimichi towns west of 31,0°E (*Propoisk (*Prupoi*)*, *Krichev*, *Luchin*, *Shklov*, *Mogilev*, *Radoml*, *Chechersk*, *Rechitsa* and *Gomel (*Homiy*)*) [15][16].

Benchmark Year	Recommended Pop.	Scholarly Low	Scholarly High	Implied Density (per km2)	Urban Pop. (Gorodishcha)	Primary Documentary & Archaeo-logical Basis
10000BC	8 000 (8,0k)	4 500	13 000	0,04	-	Yurevichi, Berdizh & Swiderian & Grensk Final Palaeolithic
5000BC	32 000 (32,0k)	20 000	48 000	0,15	-	Narva, Neman, Dnieper-Donets & Upper Dnieper Neolithic
2500BC	78 000 (78,0k)	52 000	110 000	0,38	-	Krivina pile dwellings (Asavets), Middle Dnieper & Corded Ware
1200BC	135 000 (135,0k)	95 000	185 000	0,65	-	Trzciniec-Sosnytsia & North Belarusian Bronze Age
500BC	240 000 (240,0k)	175 000	315 000	1,16	-	Milograd, Brushed Pottery & Dnieper-Dvina (>1 100 hillforts)
200AD	295 000 (295,0k)	220 000	380 000	1,42	-	Polesian Zarubintsy (Chaplin), Late Brushed Pottery & Wielbark
500AD	340 000 (340,0k)	255 000	435 000	1,64	12 000	Bancerovshchina-Tushemlya, Kolochin & Prague-Korchak
800AD	460 000 (460,0k)	350 000	580 000	2,22	35 000	Polotsk Krivichi Long Barrows, Dregovichi & Radimichi

(Continued from previous page)

Benchmark Year	Recommended Pop.	Scholarly Low	Scholarly High	Implied Density (per km2)	Urban Pop. (Gorodishcha)	Primary Documentary & Archaeo-logical Basis
900AD	530 000 (530,0k)	410 000	660 000	2,55	52 000	Early Polotsk (Gorodishche), Vitebsk, Turov & Menka crafts
1000AD	620 000 (620,0k)	490 000	760 000	2,99	78 000	Rurikid & Izyaslavichi consolidation in Polotsk & Svyatopolk in Turov
1100AD	790 000 (790,0k)	640 000	950 000	3,81	118 000	Apogee of Vseslav the Sorcerer (1044-1101AD) & Turov Yuryevichi
1200AD	980 000 (980,0k)	850 000	1 120 000	4,72	162 000	Pre-Mongol urban & kurgan peak across 188 gorodishcha
1240AD	940 000 (940,0k)	810 000	1 080 000	4,53	152 000	Batu's southern raids on Gomel (1239), Turov, Mozyr & Berestye (1240-1241)
1250AD	820 000 (820,0k)	690 000	960 000	3,95	135 000	Post-1240 trough in Polesia & Sozh; Polotsk, Vitebsk, Minsk & Novogrudok unstormed
1300AD	860 000 (860,0k)	720 000	1 000 000	4,14	148 000	Early Lithuanian-Ruthenian realm (Black Ruthenia, Polotsk & Vitebsk)

Table 1. Regional Demographic and Archaeological Data Summary (Table 1).

Prehistoric and Iron Age Foundations (10000BC - 500AD)

Final Palaeolithic and Mesolithic Recolonisation (26000BC - 5000BC)

Human settlement across the Belarusian river basins began during the interstadials of the Late Pleistocene prior to the Last Glacial Maximum (26000BC - 21000BC). The two canonical Upper Palaeolithic mammoth-hunter base camps of Belarus are Yurevichi (Yuravichy, Kalinkovichi Raion) on a high terrace of the lower Pripyat river (c. 24500BC) and Berdizh (Berdydzh, Chechersk Raion) on the Sozh river (c. 21500BC), where semi-subterranean dwellings framed with mammoth skulls, tusks and long bones of *Mammuthus primigenius* were excavated by K. M. Polikarpovich and V. D. Budko [2][17]. Following the retreat of the Poozerie (Valdai (Weichselian)) ice sheet from northern Vitebsk around 12000BC - 10000BC, Final Palaeolithic reindeer-hunting bands of the Swiderian, Lyngby and Grensk cultures colonised the sandy outwash plains and glacial lake margins of the Neman, Pripyat, Upper Dnieper, Sozh and Western Dvina valleys (*Grensk* and *Korma* on the Sozh, *Ankovichi* in the Poozerie, *Borok* on the Neman). At the 10000BC benchmark, regional carrying capacity across the tundra-parkland corridor is estimated at 8 000 inhabitants (8,0k; range 4,5k - 13,0k, 0,04 per km^2) [17].

During the Boreal and Atlantic Mesolithic (8000BC - 5000BC), rising temperatures and the spread of mixed pine-birch and broad-leaved oak-hornbeam forests supported denser semi-sedentary hunter-fisher-forager populations grouped into three archaeological provinces: the Kunda culture in the lake-rich Belarusian Poozerie (Western Dvina basin), the Janislawice (Neman) Mesolithic across the Neman and Western Bug basins and the Sozh and Upper Dnieper Mesolithic in eastern Belarus [17]. Over 400 Mesolithic camps have been catalogued along lacustrine terraces and river confluences.

Neolithic Cultures and the Krivina Peat-Bog Pile Dwellings (5000BC - 2500BC)

The Neolithic transition (5000BC - 2200BC) witnessed a four-fold expansion in regional settlement density (32 000 inhabitants at 5000BC, rising to 78 000 by 2500BC), driven by intensive lacustrine fishing, waterfowl trapping and the progressive adoption of stockbreeding and millet-barley-emmer cultivation [2][17]:

- **Narva and Usvyaty Cultures** (Vitebsk Poozerie): Famous for waterlogged pile-dwelling settlements in the Krivina peat-bog micro-region (Beshenkovichi and Senno Raions: *Asavets 2*, *Asavets 7*, *Kryvina 1*, *Aztsy*) excavated by M. M. Charniauski. Exceptional organic preservation at *Asavets* yielded wooden fish-weirs, dugout canoes, horn hoes, Baltic amber ornaments and carved bone and wooden anthropomorphic figurines [17].
- **Neman Culture** (Upper Neman and Western Bug): Characterised by pointed-bottom and flat-bottom vessels tempered with plant fibres and crushed granite (*Kamen*, *Rusakovo*, *Dobry Bor*), together with the vast subterranean Krasnoselsky flint-mining complex on the Rossa river near Volkovysk, where over 1 000 vertical shafts up to 6 - 9 m deep were sunk into Cretaceous chalk strata with antler picks to extract dark flint nodules for polished axe production across the Baltic-Dnieper region [17].
- **Dnieper-Donets Culture** (Pripyat Polesia and Lower Sozh) and Upper Dnieper Culture (Eastern Belarus): Concentrated in permanent riverine villages along the Pripyat, Yaselda, Berezina and Sozh rivers (*Litvinovichi*, *Strumen*, *Grodno-Neman*).

Bronze Age and Iron Age Hillfort Cultures (2500BC - 500AD)

During the Late Eneolithic and Bronze Age (2500BC - 700BC), Indo-European pastoralist-agriculturalist groups of the Middle Dnieper Culture (2400BC - 1500BC, a core branch of the Corded Ware (Battle-Axe) horizon, with major barrow and flat cemeteries at *Strelitsa* and *Khodosovichi* near Rogachev), Polesian and Strzyzow, North Belarusian and Trzciniec-Sosnytsia Culture (1500BC - 1000BC) cleared riverine terraces for cattle, swine and sheep-goat husbandry alongside bronze-casting workshops using imported Carpathian and Alpine copper [17]. Regional population expanded from 78 000 (78,0k) at 2500BC to 135 000 (135,0k) at 1200BC.

The Early and Middle Iron Age (700BC - 400AD) introduced local bog-iron smelting (*bloomery furnaces*) and transformed the Belarusian landscape into a dense network of over 1 100 fortified timber-and-earth hillforts (*gorodishcha*). Belarusian archaeologists divide the Iron Age into three latitudinal zones that directly prefigure early medieval Baltic-Slavic ethnocultural geography [18][19][20]:

1. **Dnieper-Dvina Culture** (700BC - 400AD, Northern Belarus and Vitebsk Poozerie): Situated across the Western Dvina and upper Dnieper basins (*Polotsk-Gorodishche*, *Vitebsk-Zamkovaya Gora*, *Lukoml*, *Novye Bateki*). Inhabitants occupied strongly fortified promontory hillforts (0,3 - 1,2 ha) with concentric earthen ramparts and timber palisades, subsisting on forest-fallow (*podseka*) agriculture, stockbreeding and ironworking [20].
2. **Brushed Pottery Culture** (*Kultura Shtrikhovannoy Keramiki*, 700BC - 400AD, Central and North-Western Belarus): Spanning the Minsk Upland and Upper Neman basin (*Malyshki*, *Zazony*, *Viazynka*, *Novogrudok*, *Slonim*). Excavations by A. G. Mitrofanov documented multi-room post-built longhouses inside ring-forts [18].
3. **Milograd Culture** (700BC - 100BC) and Zarubintsy Culture (200BC - 200AD, Southern Belarus and Pripyat and Lower Berezina-Sozh): Located across Gomel, southern Mogilev and Brest Polesia (*Milograd*, *Goroshkov*, *Chaplin*, *Mokhov*, *Voronino*, *Otverzhichi*). Unlike the small northern hillforts, Milograd and Polesian Zarubintsy communities built larger lowland enclosures (2 - 8 ha) and open agricultural villages (3 - 10 ha) equipped with iron ard-shares (*ralniki*), sickles and rotary querns [19][20].

At the 500BC benchmark, the >1 100 fortified hillforts and associated open *selyshcha* supported 240 000 inhabitants (240,0k; range 175,0k - 315,0k), rising to 295 000 (295,0k) by 200AD during the Zarubintsy and Wielbark contact phase. Across the Late Roman and Migration periods (200AD - 500AD), interaction between early Slavic agriculturalists and local Baltic substrata generated the Kiev, Kolochin (in the Sozh basin, named after the *Kolochin* hillfort near Rechitsa), Bancerovshchina-Tushemlya (central and northern Belarus, named after the *Bancerovshchina* hillfort on the Svisloch near Minsk) and Prague-Korchak (Pripyat Polesia, such as *Snyadin*, *Petrikov* and *Bereztsy*) horizons, bringing the 500AD regional population to 340 000 (340,0k; range 255,0k - 435,0k) [20][21].

Early Slavic and Kievan Rus Principalities (500AD - 1240AD)

Tribal Confederations: Krivichi, Dregovichi, Radimichi and Baltic Substrata (500AD - 980AD)

Between the 6th and 10th centuries, the *Primary Chronicle* (*Povest Vremennykh Let*) and barrow (*kurgan*) archaeology record the crystallisation of three major East Slavic tribal unions across Belarus, progressively assimilating local Baltic (East Lithuanian, Yotvingian (Sudovian) and Latgalian-Selonian) populations [2][16][21]:

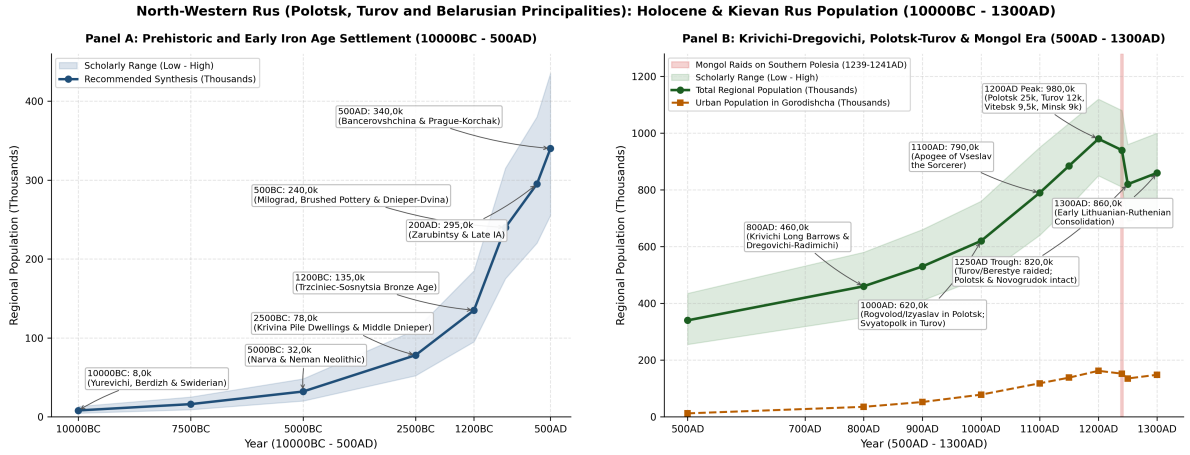


Figure 2. Holocene, Iron Age, and Kievan Rus Demographic Trajectory of Region 4 (polotsk_turov_and_belarusian_principalities) across 10000BC - 1300AD. Panel A (10000BC - 500AD) traces the expansion from Final Palaeolithic mammoth- and reindeer-hunter camps (*Yurevichi*, *Berdizh*, *Grensk*; 8,0k at 10000BC) through the Narva/Neman Neolithic (32,0k at 5000BC), the Krivina pile dwellings and Middle Dnieper Bronze Age (78,0k at 2500BC; 135,0k at 1200BC), and the Iron Age hillfort cultures (Milograd, Brushed Pottery, Dnieper-Dvina; 240,0k at 500BC and 295,0k at 200AD). Panel B (500AD - 1300AD) traces the Krivichi-Dregovichi-Radimichi colonisation (460,0k at 800AD), Rurikid consolidation (620,0k at 1000AD), the apogee of Vseslav the Sorcerer (790,0k at 1100AD), the 1200AD pre-Mongol peak (980,0k), the 1239-1241AD southern Polesian raids (820,0k trough at 1250AD), and early Lithuanian-Ruthenian consolidation (860,0k at 1300AD).

- **The Polotsk-Smolensk Krivichi** (*Polochane*): Settled along the Western Dvina, Polota, Drissa, Ulla, Kasplya and Upper Berezina rivers. Archaeologically marked by elongated Long Barrows (*Dlinnye Kurgany*, 6th - 9th c.) containing multiple cremation burials, followed in the 9th - 11th c. by hemispherical barrows with distinctive bronze bracelet-like temple rings (*brasletoobraznye visochnye koltsa*) [1][21].
- **The Dregovichi** (*Drygavichy*): Occupying the Pripyat basin north to the Minsk Upland (overlapping with the Krivichi along the Svisloch-Berezina watershed at *Zaslavl*, *Minsk*, *Logoisk* and *Borisov*). Dregovichi burial mounds are distinguished by large granulated silver and bronze bead temple rings (*krupnozernistye businnye koltsa*) and timber mortuary chambers (*domoviny*) [10][16].
- **The Radimichi**: Settled along the Sozh, Pronya, Iput, Besed and Ostyor rivers in eastern Belarus (with major tribal hillforts at *Radoml*, *Propoisk (*Prupoi*)* and *Gomel (*Homiy*)*), identified by seven-rayed bronze temple rings (*semiluchevye visochnye koltsa*) [15][16].
- **Yotvingians** (*Yatvyagi*), Lithuanians (*Litva*) and Latgalians and Selonians: Occupying the Upper Neman (*Grodno*, *Novogrudok*, *Volkovysk*, *Slonim*, *Lida*, *Krevo*) and Lower Western Dvina (*Braslav*, *Jersika*, *Koknese*, *Asote*) borderlands, where mixed Slavic-Baltic stone-ring barrows (*kamennye mogilniki*) attest to bilingual frontier symbiosis [8][13][14].

Regional population expanded from 460 000 (460,0k) at 800AD to 530 000 (530,0k) at 900AD and 620 000 (620,0k) at 1000AD, comparing directly with 310 000 in *Novgorod-Pskov* and 220 000 in *Eastern Smolensk* within European Russia.

The Izyaslavichi Principalities of the Polotsk Land (1001AD - 1241AD)

Unlike the southern and eastern principalities of Kievan Rus, the Principality of Polotsk (polity ID 2460, 1001-1241AD, 55,49°N, 28,80°E) established permanent dynastic autonomy under the Izyaslavichi (*Rogvolodovichi*), descendants of Prince Rogvolod (d. 978AD), his daughter Rogneda and her son by Vladimir the Great, Izyaslav Vladimirovich (d. 1001AD) [5][6]. Under Bryachislav Izyaslavich (1003-1044AD) and Vseslav Bryachislavich the Sorcerer (1044-1101AD), Polotsk controlled the entire Western Dvina trade artery from the Vitebsk-Smolensk portages down to the Gulf of Riga, erecting the monumental seven-domed Cathedral of St. Sophia in Polotsk (c. 1050s) as a direct architectural and ideological counterpart to Kiev and Novgorod [5][7].

Stratigraphic excavations in Polotsk (*Polatsk*) by G. V. Shtykhov (1975, 1992) and historical-topographical reconstructions by S. V. Tarasov (1998) reveal one of the largest urban agglomerations of pre-Mongol Eastern Europe [5][7]:

- Initial Hillfort (*Gorodishche*, 9th - 10th c., 2,5 ha) on the Polota river, succeeded in the late 10th century by the Upper Castle (*Verkhny Zamok* (*Detinets*), 11,0 ha) at the Polota-Dvina confluence, housing St. Sophia Cathedral and the princely court.
- The Great Posad (*Velikiy Posad*, 11th - 12th c.), Lower Castle (*Nizhny Zamok* (*Zapadnaya Dvina Posad*), 63,0 ha), Zapolotye (west of the Polota), Ostrov (island quarter), Euphrosyne Suburb (around the 1128AD Saviour-Euphrosyne Monastery founded by St. Euphrosyne of Polotsk, with the frescoed Church of the Transfiguration built by Master Ioann) and the Belchitsy princely residence on the left bank of the Dvina.
- By 1200AD, the contiguous settled footprint of Polotsk reached 180 - 200 ha (190,0 ha central synthesis), supporting 22 000 - 28 000 inhabitants (25 000 recommended) across 3 800 - 4 200 urban homesteads (*dvory*), surrounded by satellite towns at Lukoml (3 500), Logoisk (*Logozhsk*) (3 500), Zaslavl (*Izyaslavl*) (20,0 ha, 4 000), Braslav (*Bryachislavl*) (3 000), Usvyaty (2 800) and Svisloch-on-Berezina (2 500) [1][4][7].

After the death of Vseslav the Sorcerer in 1101AD, the Polotsk Land devolved into six appanage principalities (*udely*), each stimulating secondary urban growth:

1. **Principality of Minsk** (polity ID 2761, 1102-1327AD, 53,90°N, 27,56°E): Founded by Gleb Vseslavich (d. 1119AD). E. M. Zagorulsky's excavations demonstrated that early *Mensk* originated in the 10th-11th centuries at the Menka archaeological complex (Gorodishche village, 16 km west of modern Minsk on the Menka river, comprising a 1,5 ha two-part ramparted *gorodishche* and a 40,0 ha open *selyshche* peaking at 2 400 inhabitants in 1000AD). Following the 1067AD Battle on the Nemiga River recorded in the *Primary Chronicle* and the *Tale of Igor's Campaign*, the fortress was transferred to the Minsk Zamchishche at the Svisloch-Nemiga confluence (3,0 ha timber-laced rampart enclosing an unfinished 11th-c. dressed-limestone church, surrounded by 35,0 ha of waterlogged *posad* log streets), reaching 38,0 ha and 9 000 inhabitants (8 000 - 10 500) at 1200AD [2][9].
2. **Principality of Vitebsk** (polity ID 2762, 1102-1321AD, 55,19°N, 30,20°E): Held by Svyatoslav and Davyd Vseslavich at the Vitba-Western Dvina junction. Comprising the *Verkhny Zamok* (Zamkovaya Gora), *Nizhny Zamok* and *Uzgorier* posads (50,0 ha total), Vitebsk housed the 1140s Church of the Annunciation (built of alternating brick *plinthos* and dressed dolomite blocks), yielded the late 13th-century birch-bark letter of Stepan (Nezhata owed 6 grivnas) and supported 9 500 inhabitants (8 000 - 11 000) at 1200AD, alongside the Dnieper portage fortress of Orsha (*Rsha*, 1067AD, 22,0 ha, 4 000 inhabitants) and Kopys (1059AD, 2 800) [1][6].
3. **Principality of Drutsk** (polity ID 2760, 1102-1251AD, 54,34°N, 29,75°E): Ruled by Rogvolod Borisovich (fl. 1144-1171AD, patron of the 1178AD *Drutsk Gospel* and the inscribed *Boris Stones* on the

Dvina). Excavations by L. V. Alekseev at Drutsk on the upper Drut portage documented a 2,5 ha detinets, a 9,5 ha *okolny gorod* and 23,0 ha of posads (35,0 ha total, 4 500 inhabitants) [6].

4. **Principality of Borisov** (polity ID 2759, 1102-1301AD, 54,23°N, 28,52°E): Founded on the Berezina river in 1102AD by Prince Boris Vseslavich at Stary Borisov (18,0 ha complex, 3 500 inhabitants) [1].
5. **Tributary Principalities of Jersika** (polity ID 2459, 1001-1210AD, 56,25°N, 26,19°E) and Koknese (polity ID 3106, 1191-1210AD, 56,64°N, 25,42°E): Controlling the lower Western Dvina in Latgale and Selonia. According to Henry of Livonia's *Chronicon Livoniae*, Orthodox Prince Visvaldis (Vsevolod) of Jersika ruled from the 1,5 ha cliff-top hillfort and 12,0 ha posad of *Jersika (Gerzike)* (2 500 inhabitants) until Bishop Albert of Riga and the Livonian Brothers of the Sword stormed and burned its Orthodox churches in 1209AD, while Prince Vyachko (Vetseke) of Koknese held the Dvina-Perse promontory fortress of *Koknese (Kukenois)* (10,0 ha, 2 000 inhabitants) until burning his own castle in 1208AD [6][8].

The Turov-Pinsk Principalities and Berestye Land (1133AD - 1300AD)

Across southern Belarus, the Pripyat river and its ten navigable tributaries formed the Land of Turov and Pinsk. First attested under Prince Tury in 980AD and ruled in the 11th century as the senior throne of the Kievan heir apparent (Svyatopolk Vladimirovich, Izyaslav Yaroslavich, Svyatopolk Izyaslavich), Turov (polity IDs 2855, 3356, 3471, 1133-1341AD, 52,07°N, 27,74°E) secured permanent dynastic independence under Vyacheslav Vladimirovich and Yuri Yaropolkovich (d. c. 1167AD) after withstanding a ten-week coalition siege in 1158AD [3][10].

P. F. Lysenko's excavations (1961-1968, 1992-2003) at Turov (*Turau*) on the Pripyat documented the 2,5 ha *Zamkova Hora* detinets (containing the brick foundations of the 1170s Cathedral of the Dormition, 29,3 x 17,9 m, seat of the celebrated homilist Bishop Cyril of Turov, c. 1130-1182AD, together with four lead patriarchal and episcopal seals and carved Ovruch slate sarcophagi), the 12,5 ha *Okolny Gorod* and 30,0 ha of alluvial posads (45,0 ha total complex, 12 000 inhabitants at 1200AD), flanked by Mozyr (16,5 ha, 3 500), David-Gorodok (14,0 ha, 3 000), Rogachev (14,0 ha, 3 000), Bragin (11,5 ha, 2 800) and Strelsk (1 800) [3].

In the late 12th century, Yuri Yaropolkovich's sons partitioned the Polesian basin into multiple appanages:

- **Principality of Pinsk** (polity ID 3051, 1175-1321AD, 52,11°N, 26,10°E): Located at the Pina-Strumen-Pripyat confluence (*Pinsk (*Pinesk*)*, 1097AD), where a 3,2 ha detinets, 10,8 ha *okolny gorod* and 24,0 ha posad (38,0 ha total) supported 8 000 inhabitants (6 500 - 9 500) at 1200AD, alongside *Nobel* (1 600) and *Pogost* (1 400) [10].
- **Principality of Kletsk** (polity ID 2920, 1148-1401AD, 53,06°N, 26,63°E) and Slutsk (*Sluchesk*, 1116AD, 53,03°N, 27,56°E): Anchoring the fertile northern Polesian forest-steppe fringe (loess islands of the Kletsk-Slutsk-Kopyl-Nesvizh ridge). Slutsk on the Sluch river (25,0 ha, 5 000 inhabitants), Kletsk on the Lan river (18,0 ha, 3 500 inhabitants) and Kopyl (10,0 ha, 2 500 inhabitants) formed major agricultural and craft centres [10].
- **Principality of Dubrovytsia** (polity ID 3050, 1175-1301AD, 51,57°N, 26,56°E): Seat of Prince Gleb Yuryevich (fl. 1183-1190AD, who fought at the Battle of the Kayala in 1185AD) on the Horyn river (18,0 ha, 3 500 inhabitants), linked with Stepan (2 200) and Vysotsk (1 250) [10].
- **The Berestye Land** (*Beresteyskaya Zemlya*): On the western edge of Polesia at the Western Bug-Mukhavets confluence, Berestye (*Brest*, 1019AD) served as the strategic customs and military gateway between Turov, Volhynia, Mazovia and Yatvingia. Between 1969 and 1981, P. F. Lysenko excavated 1 800 m^2 of the waterlogged *Hospital Island* detinets (4 - 7 m cultural layer), uncovering 12 chronological tiers of pine log-corduroy streets (11th - 13th c.) and over 220 log cabins preserved up to 12 logs (1,8 m) in height complete with doorways, windows and oak shingles [11]. Berestye

covered 34,0 ha (7 500 inhabitants) at 1200AD, guarded to the north by Kamenets (1276AD), where Volhynian Prince Vladimir Vasilkovich and architect Aleksa erected the 30 m brick Tower of Kamenets (*Belaya Vezha*, 3 000 inhabitants by 1300AD), Kobrin (2 000) and Zditov (1 650) [11].

Black Ruthenia (Upper Neman) and the Western Smolensk-Sozh Corridor (1128AD - 1241AD)

Along the Upper Neman river, Slavic colonisation by Dregovichi, Volhynians and Krivichi into Yotvingian and Lithuanian territory created the principalities of Black Ruthenia (*Chornaya Rus*) [12][13][14]:

- **Principality of Grodno** (*Goroden*) (polity ID 2826, 1128-1201AD, 53,69°N, 23,83°E): Ruled by Prince Vsevolodko Davydovich (d. 1141AD, husband of Vladimir Monomakh's daughter Agafia) and his sons Boris and Gleb. Excavated by J. Jodkowski and N. N. Voronin (1954), Grodno (35,0 ha, 8 500 inhabitants) developed an autonomous school of monumental masonry exemplified by the Kalozha Church of SS. Boris and Gleb (c. 1180s), the Lower Church and the Princely Palace on *Staraya Gora*, featuring walls inlaid with polished granite boulders and glazed polychrome ceramic crosses [12][14].
- **Principality of Novogrudok** (polity ID 3183, 1201-1241AD, 53,60°N, 25,82°E): Situated on the high *Zamkovaya Gora* and *Maly Zamok* hills (323 m elevation). F. D. Gurevich's excavations (1956-1985) revealed an exceptionally wealthy 12th-13th-century boyar and goldsmith quarter on the *Okolny Gorod* containing two-storey timber mansions equipped with glazed window glass, frescoed plaster interiors, Byzantine amphorae, Syrian enamelled glass goblets and gold and silver casting matrices (32,0 ha total complex, 7 500 inhabitants at 1200AD, rising to 9 000 by 1300AD as Mindaugas's early capital) [13].
- **Principality of Volkovysk** (polity ID 3184, 1201-1241AD, 53,16°N, 24,45°E): Comprising the fortified hills of *Shvedskaya Gora*, *Zamchishche* and *Muravelnik* (24,0 ha, 5 000 inhabitants per Ya. G. Zverugo 1989), alongside Slonim (*Uslonim*, 12,0 ha, 3 500), Turiisk-on-Neman (1 600) and Zditovo-on-Molchad (1 500) [14].
- **Principality of Mstislavl** (polity ID 3079, 1181-1198AD, 54,02°N, 31,73°E) and the Upper Dnieper-Sozh Corridor: Founded in 1135AD by Smolensk Prince Rostislav Mstislavich (d. 1167AD) on the Vikhra river and named after his father Mstislav the Great. Excavations by L. V. Alekseev and M. A. Tkachev on *Zamkovaya Gora* (26,0 ha total complex, 5 000 inhabitants at 1200AD) recovered a birch-bark fragment, prince's seal and rich 12th-century craft layers. Together with Krichev (1136AD, 3 000), Propoisk (*Prupoi*) (1136AD, 2 800), Rechitsa (2 500), Chechersk (1159AD, 2 000), Luchin (1 850), Shklov (1 750), Mogilev (1 950) and Gomel (*Homiy*) (1142AD, 30,0 ha, 6 500 inhabitants on the lower Sozh), this eastern Belarusian corridor held 130 000 inhabitants in 1200AD (separate from the 340 000 inhabitants of eastern Smolensk east of 31,0°E in European Russia Cluster 4) [15].

Sub-Regional Devolved Principality Cluster (Polity IDs)	800AD Pop.	1000AD Pop.	1100AD Pop.	1200AD Peak	1250AD Trough	1300AD Pop.	Gorodishcha (N)
1. Polotsk, Vitebsk, Jersika & Koknese (2460, 2762, 2459, 3106)	135 000	180 000	230 000	285 000	260 000	270 000	46 (5 >10 ha)
2. Minsk, Drutsk & Borisov (2761, 2760, 2759)	65 000	90 000	115 000	145 000	130 000	138 000	34 (4 >10 ha)
3. Turov, Pinsk, Kletsk, Dubrovytsia & Berestye (2855, 3051, 2920, 3050)	130 000	170 000	215 000	270 000	195 000	205 000	48 (7 >10 ha)
4. Black Ruthenia: Grodno, Novogrudok & Volkovysk (2826, 3183, 3184)	70 000	95 000	120 000	150 000	145 000	155 000	32 (4 >10 ha)
5. Mstislavl, W. Smolensk & Upper Sozh (3079, west of 31,0°E)	60 000	85 000	110 000	130 000	90 000	92 000	28 (3 >10 ha)
Total Region 4 (207 600 km2)	460 000	620 000	790 000	980 000	820 000	860 000	188 (23 >10 ha)

Table 2. Regional Demographic and Archaeological Data Summary (Table 2).

Mongol Raids and Lithuanian-Ruthenian Era (1238AD - 1300AD)

Asymmetric Impact of the 1239-1241AD Campaigns: Southern Polesia vs Northern Polotsk and Black Ruthenia

A defining structural feature of North-Western Rus demographic history is the sharp north-south divergence during Batu Khan's Western Campaign (1237-1241AD) [1][2][3]:

1. **Southern Polesian and Lower Sozh Destruction Layers** (1239-1241AD): During the autumn of 1239AD, Mangu's and Batu's columns operating against the Chernigov Principality stormed Gomel (*Homiy*), where excavations on the Sozh promontory revealed burned timber fortifications, weapons caches and skeletal remains in 1239AD destruction horizons (6 500 in 1200AD falling to 3 500 in 1250AD). In the winter of 1240-1241AD, following the fall of Kiev, Mongol detachments advancing

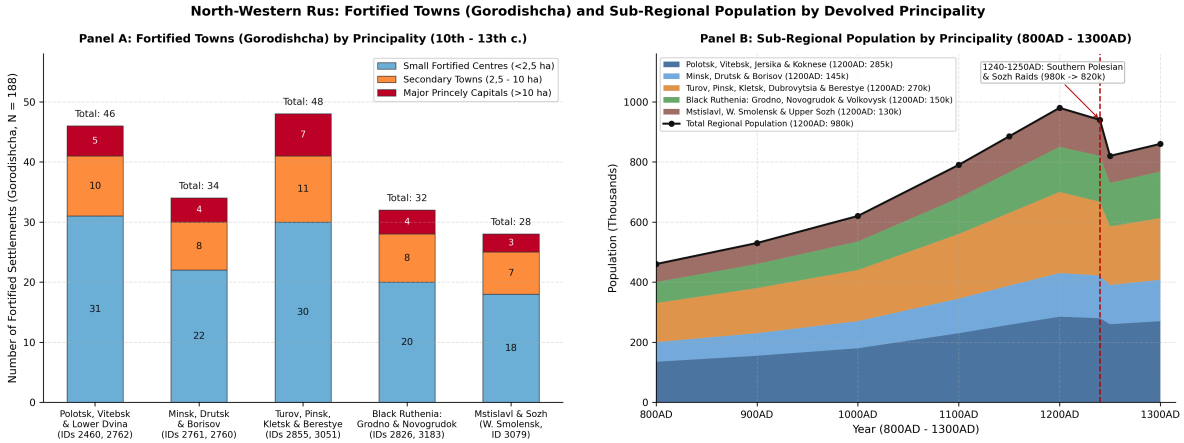


Figure 3. Fortified Towns (*Gorodishcha*) and Sub-Regional Population Breakdown across the Five Devolved Principality Clusters (800AD - 1300AD). Panel A classifies the 188 archaeologically attested 10th-13th century fortified centres into Small Fortified Centres (<2,5 ha, N = 121), Secondary Towns (2,5 - 10 ha, N = 44), and Major Princely Capitals (>10 ha, N = 23) per Kuza (1989, 1996) and Shtykhov (1992). Panel B traces the sub-regional demographic trajectory from 800AD (460,0k) to the 1200AD pre-Mongol peak (980,0k), the asymmetric 1240-1250AD contraction in Turov-Pinsk and the Sozh basin (820,0k), and the 1300AD consolidation (860,0k).

westwards toward Vladimir-in-Volhynia and Poland struck the southern Pripyat corridor, sacking Turov (12 000 -> 7 000), Mozyr (3 500 -> 2 100), Bragin (2 800 -> 1 500), Dubrovytsia (3 500 -> 2 200) and Berestye (7 500 -> 5 200), where the Hypatian Chronicle records unburied corpses lying across the fields between Vladimir and Berestye in early 1241AD [3][11]. Subsequent punitive expeditions led by the Mongol general Burundai (1258-1259AD) passed through Polesia and compelled Volhynian and local forces to dismantle fortifications along the southern frontier.

2. **Unstormed Continuity in Polotsk, Vitebsk, Minsk and Black Ruthenia** (1238-1300AD): Conversely, protected by the dense marshy forests of the Berezina-Pripyat Polesian lowland and the upper Dnieper watershed, Polotsk (25 000 -> 22 000 -> 23 000), Vitebsk (9 500 -> 8 800 -> 9 500), Minsk (9 000 -> 8 200 -> 8 800), Drutsk (4 500 -> 4 100 -> 4 300), Borisov (3 500 -> 3 200 -> 3 400), Grodno (8 500 -> 8 000 -> 8 500), Novogrudok (7 500 -> 8 200 -> 9 000) and Volkovysk (5 000 -> 4 700 -> 5 100) were never besieged or stormed during the 1237-1241AD invasion [1][7][13]. While trade disruptions along the Dnieper-Black Sea route and Livonian crusader pressure on the lower Dvina (*Jersika* and *Koknese* having fallen to Riga in 1208-1209AD) trimmed Polotsk's long-distance commerce, northern and western Belarus absorbed artisan and boyar refugees fleeing the devastated Middle Dnieper and southern Rus principalities.

Transition into the Early Lithuanian-Ruthenian Realm (1240s - 1300AD)

In the immediate aftermath of the 1238-1241AD Mongol campaigns, the unstormed urban network and boyar *veche* institutions of Black Ruthenia and the Polotsk Land catalyzed the formation of the early Grand Duchy of Lithuania (1240s - 1300AD) [13][22][23]:

- **Black Ruthenia under Mindaugas and Vaišvilkas** (1240s - 1268AD): By the mid-1240s, Novogrudok (3183), Grodno (2826), Volkovysk (3184) and Slonim formed the military, fiscal and artisanal heartland of Mindaugas (d. 1263AD) and his Orthodox son Vaišvilkas (*Voishalk*, 1264-1267AD), who

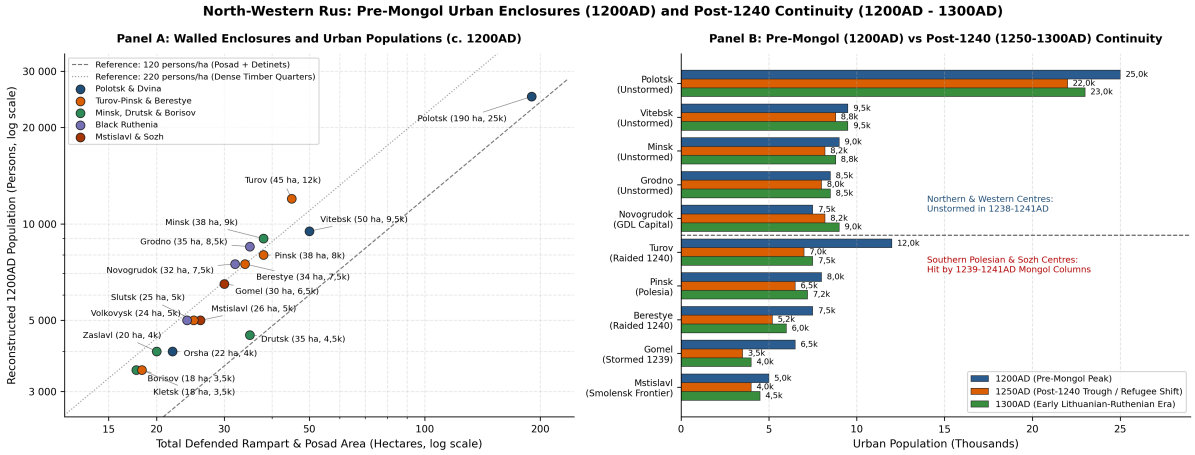


Figure 4. Pre-Mongol Urban Enclosures (c. 1200AD) and Post-1240 Urban Continuity (1200AD - 1300AD). Panel A plots total defended rampart and *posad* area (ha, log scale) against reconstructed 1200AD urban population (persons, log scale) for the 17 principal towns of Region 4 against reference density lines (120 and 220 persons/ha). Panel B compares urban populations at 1200AD (Pre-Mongol Peak), 1250AD (Post-1240 Trough), and 1300AD (Early Lithuanian-Ruthenian Consolidation) between the unstormed northern/western capitals (*Polotsk*, *Vitebsk*, *Minsk*, *Grodno*, *Novogrudok*) and the raid-impacted southern Polesian and Sozh centres (*Turov*, *Pinsk*, *Berestye*, *Gomel*, *Mstislavl*).

founded the Lavrishevo Monastery on the Neman near Novogrudok. Archaeological strata in late 13th-century Novogrudok show continuous expansion of its boyar and craft quarters from 7 500 inhabitants in 1200AD to 8 200 in 1250AD and 9 000 in 1300AD [13].

- **Polotsk and Vitebsk under Tautvilas, Gerdenis and Iziaslav** (1252-1300AD): Through dynastic marriages and *veche* treaties preserving autonomous Polotsk law (*starina*), Mindaugas's nephew Tautvilas (*Tovtivil*, Prince of Polotsk c. 1252-1263AD), Gerdenis (1264-1267AD, signatory of the 1264AD Polotsk-Vitebsk trade treaty with Riga and Gotland) and Prince Константин Bezrukii (c. 1270s-1290s) maintained Polotsk (23 000 in 1300AD) and Vitebsk (9 500 in 1300AD) as self-governing principalities allied with Lithuania against the Livonian Order and the Golden Horde [1][7][22].
- **Regional Demographic Balance at the 1300AD Bookend**: Across Region 4 as a whole, total population dipped by -16,3% from its 1200AD pre-Mongol peak of 980 000 (980,0k) to 820 000 (820,0k) at the 1250AD post-conquest trough, before recovering to 860 000 (860,0k; range 720,0k - 1 000,0k, 4,14 per km^2) by 1300AD (Polotsk-Vitebsk-Minsk-Drutsk-Borisov 408,0k, Turov-Pinsk-Kletsk-Dubrovytsia-Berestye 205,0k, Black Ruthenia 155,0k, Mstislavl and Upper Sozh 92,0k), providing the intact demographic foundation of the Lithuanian-Ruthenian state at the close of the 13th century [22][23].

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